LG Chem	SAFETY DATA SHEET	Version: 12th
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		Revision date: 2017-11-22
	METHYL ACRYLATE	Change List:

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1. IDENTIFICATION

A. Product name

- METHYL ACRYLATE

B. Recommended use and restriction on use

- General use : Acrylic Fiber, Paper Coating, synthetic textiles, leather, etc
- Restriction on use : Do not use non-recommended purposes

C. Manufacturer / Supplier / Distributor information

○ Manufacturer information

- Company name : LG Chem
- Address : 55, Yeosusandan 2-ro, Yeosu-Si, Jeonnam, 555-718, Korea
- Dept. : IPA Production Team
- Telephone number : 061-680-1331
- Emergency telephone number : 061-680-1333
- Fax number : 061-680-6044
- E-mail address :

○ Supplier/Distributer information

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2. HAZARD IDENTIFICATION

A. GHS Classification

- Flammable liquids : Category2
- Acute toxicity (oral) : Category4
- Acute toxicity (dermal) : Category4
- Acute toxicity (inhalation: vapor) : Category3
- Skin corrosion/irritation : Category2
- Skin sensitization : Category1
- Specific target organ toxicity(Single exposure) : Category3(Respiratory tract irritation)

B. GHS label elements

○ Hazard symbols

○ Signal words

- Danger

○ Hazard statements

- H225 Highly flammable liquid and vapour
- H302 Harmful if swallowed
- H312 Harmful in contact with skin

- H315 Causes skin irritation
- H317 May cause an allergic skin reaction
- H331 Toxic if inhaled
- H335 May cause respiratory irritation.

○ Precautionary statements

1) Prevention
- P210 Keep away from heat/sparks/open flames/hot surfaces. No smoking.
- P233 Keep container tightly closed.
- P240 Ground/bond container and receiving equipment.
- P241 Use explosion-proof electrical/ventilating/lighting/equipment.
- P242 Use only non-sparking tools. Flammable liquids (chapter 2.6) 1, 2, 3
- P243 Take precautionary measures against static discharge.
- P261 Avoid breathing gas/mist/vapours/spray.
- P264 Wash hands thoroughly after handling.
- P270 Do not eat, drink or smoke when using this product.
- P271 Use only outdoors or in a well-ventilated area.
- P272 Contaminated work clothing should not be allowed out of the workplace.
- P280 Wear protective gloves/protective clothing/eye protection/face protection.

2) Response
- P301+P312 IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
- P302+P352 IF ON SKIN: Wash with plenty of soap and water.
- P303+P361+P353 IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- P304+P340 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
- P311 Call a POISON CENTER or doctor/physician.
- P312 Call a POISON CENTER or doctor/physician if you feel unwell.
- P321 Specific treatment
- P330 Rinse mouth.
- P332+P313 If skin irritation occurs: Get medical advice/attention.
- P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
- P362 Take off contaminated clothing and wash before reuse.
- P363 Wash contaminated clothing before reuse.
- P370+P378 In case of fire: Use Suitable extinguishing media for extinction(Refer Section MSDS 5).

3) Storage
- P403+P233 Store in a well-ventilated place. Keep container tightly closed.
- P403+P235 Store in a well-ventilated place. Keep cool.
- P405 Store locked up.

4) Disposal
- P501 Dispose of contents/container in accordance with local/regional/national/international regulation

C. Other hazards which do not result in classification : (NFPA Classification)

○ NFPA grade (0 ~ 4 level)
- Health : 3 , Flammability : 3, Reactivity : 2

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3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	Trade names and Synonyms	CAS No.	Content(%)
2-Propenoic acid methyl ester	Methyl acrylate	96-33-3	100

4. FIRST AID MEASURES

A. Eye contact
- Do not rub your eyes.
- Immediately flush eyes with plenty of water for at least 15 minutes and call a doctor/physician.
- Get medical attention immediately.

B. Skin contact

- Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes.
- Laundering enough contaminated clothing before reuse.
- Get medical attention immediately.
- Go to the hospital immediately if symptoms(flare, irritate) occur.
- Prevent the spread of the skin.
- Wash thoroughly after handling.

C. Inhalation contact

- When exposed to large amounts of steam and mist, move to fresh air.
- Take specific treatment if needed.
- Get medical attention immediately.
- If breathing is stopped or irregular, give artificial respiration and supply oxygen.
- Take the doctor's examination.

D. Ingestion contact

- Please be advised by doctor whether induction of vomit is demanded or not.
- Rinse your mouth with water immediately.
- Get medical attention immediately.

E. Delayed and immediate effects and also chronic effects from short and long term exposure

- Not available

F. Notes to physician

- Notify medical personnel of contaminated situations and have them take appropriate protective measures.

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5. FIREFIGHTING MEASURES

A. Suitable (Unsuitable) extinguishing media

- Dry chemical, carbon dioxide, regular foam extinguishing agent, spray
- Avoid use of water jet for extinguishing

B. Specific hazards arising from the chemical

- Not available

C. Special protective actions for firefighters

- Move containers from fire area, if you can do without the risk.
- Cool containers with water until well after fire is out.
- Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Notify your local firestation and inform the location of the fire and characteristics hazard.
- Using a unattended and water devices in case of large fire and leave alone to burn if you do not imperative.
- Keep containers cool with water spray.
- Vapor or gas is burned at distant ignition sources can be spread quickly.
- Due to the extremely low flash point, irrigating fire extinguishing may be less effective when put out a fire.

6. ACCIDENTAL RELEASE MEASURES

A. Personal precautions, protective equipment and emergency procedures

- Ventilate closed spaces before entering.
- Must work against the wind, let the upwind people to evacuate.
- Remove all sources of ignition.
- Do not direct water at spill or source of leak.
- Avoid skin contact and inhalation.

B. Environmental precautions

- Prevent runoff and contact with waterways, drains or sewers.
- If large amounts have been spilled, inform the relevant authorities.

C. Methods and materials for containment and cleaning up

- Large spill : Stay upwind and keep out of low areas. Dike for later disposal.
- Notification to central government, local government. When emissions at least of the standard amount
- Dispose of waste in accordance with local regulation.
- Appropriate container for disposal of spilled material collected.
- Small leak: sand or other non-combustible material, please let use absorption.
- Wipe off the solvent.
- Dike for later disposal.
- Do not use plastic containers.

7. HANDLING AND STORAGE

A. Precautions for safe handling

- Since emptied containers retain product residue(vapor, liquid, solid) follow all MSDS and label warnings even after container is emptied.
- Refer to Engineering controls and personal protective equipment.
- Do not handle until all safety precautions have been read and understood.
- Operators should wear antistatic footwear and clothing.
- Do not inhale the steam prolonged or repeated.
- Avoid contact with heat, sparks, flame or other ignition sources.

B. Conditions for safe storage, including any incompatibilities

- Do not use damaged containers.
- Do not apply direct heat.
- Do not apply any physical shock to container.
- No open fire.
- Prevent static electricity and keep away from combustible materials or heat sources.
- Collected them in sealed containers.
- Do not eat, drink or smoke when using this product.

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8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Exposure limits

○ ACGIH TLV

- [2-Propenoic acid methyl ester] : TWA, 2 ppm (7 mg/m3) Skin SEN

○ OSHA PEL

- [2-Propenoic acid methyl ester]:10ppm 35mg/m3

B. Engineering controls

- A system of local and/or general exhaust is recommended to keep employee exposures above the Exposure Limits. Local exhaust ventilation is generally preferred because it can control the emissions of the contaminant at its source, preventing dispersion of it into the general work area. The use of local exhaust ventilation is recommended to control emissions near the source.

C. Individual protection measures, such as personal protective equipment

○ Respiratory protection

- Under conditions of frequent use or heavy exposure, Respiratory protection may be needed.
- Respiratory protection is ranked in order from minimum to maximum.
- Consider warning properties before use.
- Any chemical cartridge respirator with organic vapor cartridge(s).
- Any chemical cartridge respirator with a full facepiece and organic vaporcartridge(s).
- Any air-purifying respirator with a full facepiece and an organic vapor canister.
- For Unknown Concentration or Immediately Dangerous to Life or Health : Any supplied-air respirator with full facepiece and operated in a pressure-demand or other positive-pressure mode in combination with a separate escape supply. Any self-contained breathing apparatus with a full facepiece.

○ Eye protection

- Wear primary eye protection such as splash resistant safety goggles with a secondary protection face shield.
- Provide an emergency eye wash station and quick drench shower in the immediate work area.

○ Hand protection

- Wear appropriate chemical resistant glove.

○ Skin protection

- Wear appropriate chemical resistant protective clothing.

○ **Others**

- Not available

9. PHYSICAL AND CHEMICAL PROPERTIES

A. Appearance	
- Appearance	Liquid
- Color	Colorless
B. Odor	Sour smell
C. Odor threshold	Water : 0.0021mg/L
D. pH	Not available
E. Melting point/Freezing point	-76.5 ℃
F. Initial Boiling Point/Boiling Ranges	80.1 ℃
G. Flash point	-2.8 ℃
H. Evaporation rate	Not available
I. Flammability(solid, gas)	Highly Flammable liquid
J. Upper/Lower Flammability or explosive limits	14.5 / 2.1vol%
K. Vapour pressure	90h㎩ (20℃)
L. Solubility	60g/L (20℃)
M. Vapour density	4.42 (air=1)
N. Specific gravity(Relative density)	0.95g/㎤ (20℃)
O. Partition coefficient of n-octanol/water	LogKow = 0.739 (25℃)
P. Autoignition temperature	468 ℃
Q. Decomposition temperature	Not available
R. Viscosity	0.472 m㎩ (25℃)
S. Molecular weight	86.09

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10. STABILITY AND REACTIVITY

A. Chemical Stability

- This material is stable under recommended storage and handling conditions.

B. Possibility of hazardous reactions

- Cylinders exposed to fire may vent and release flammable gas.

C. Conditions to avoid

- Avoid contact with incompatible materials and condition.
- Avoid : Accumulation of electrostatic charges, Heating, Flames and hot surfaces
- Avoid contact with heat, sparks, flame or other ignition sources.

D. Incompatible materials

- Not available

E. Hazardous decomposition products

- May emit flammable vapour if involved in fire.

11. TOXICOLOGICAL INFORMATION

A. Information on the likely routes of exposure

○ **(Respiratory tracts)**

- May cause respiratory irritation.

○ **(Oral)**

- Harmful if swallowed

○ **(Eye·Skin)**

- Causes skin irritation
- May cause an allergic skin reaction

B. Delayed and immediate effects and also chronic effects from short and long term exposure

O Acute toxicity

* Oral
  - [2-Propenoic acid methyl ester] : LD50 = 768 ㎎/㎏ Rat
* Dermal
  - [2-Propenoic acid methyl ester] : LD50 = 1250 ㎎/㎏ Rabbit
* Inhalation
  - [2-Propenoic acid methyl ester] : LC50 = 6.5 mg/L Rat

O Skin corrosion/irritation
- Causes skin irritation (Rabbit)

O Serious eye damage/irritation
- Causes eye corrosion (Rabbit)

O Respiratory sensitization
- Not available

O Skin sensitization
- Positive (Mouse)

O Carcinogenicity

* NOAEC
  - [2-Propenoic acid methyl ester] : >=0.519mg/L air
* IARC
  - [2-Propenoic acid methyl ester] : Group 3
* OSHA
  - Not available
* ACGIH
  - [2-Propenoic acid methyl ester] : A4
* NTP
  - Not available
* EU CLP
  - Not available

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O Germ cell mutagenicity
- NOAEC=75ppm, negative

O Reproductive toxicity
- Not available

O STOT-single exposure
- 90 day repeated dose oral toxidity test : NOAEL=0.082mg/L air

O STOT-repeated exposure
- Not classified

O Aspiration hazard
- Not available

12. ECOLOGICAL INFORMATION

A. Ecotoxicity

O Fish
- [2-Propenoic acid methyl ester] : LC50 1.1 ㎎/ℓ 96 hr Sheepshead Minnow (CERI/NITE Hazard Assessment Report (preliminary version), 2006)

O Crustaceans
- [2-Propenoic acid methyl ester] : EC50 2.6 ㎎/ℓ 48 hr

O Algae
- [2-Propenoic acid methyl ester] : ErC50 3.55 ㎎/ℓ 72 hr

B. Persistence and degradability

O Persistence
- [2-Propenoic acid methyl ester] : logPow = 0.74 , BCF = 3.162

O Degradability
- [2-Propenoic acid methyl ester] : Hydrolysis pH7 = over 28days, pH11 = 1.8hr
  Photolysis half life = 40.9hr

C. Bioaccumulative potential

○ Bioaccumulative potential

- [2-Propenoic acid methyl ester] : BCF = 2.4

○ Biodegration

- [2-Propenoic acid methyl ester] : TOC: 100 (%) (NITE: existing chemical safety inspections data)

D. Mobility in soil

- [2-Propenoic acid methyl ester] : Koc = 6.04, low potency of mobility to soil

E. Other adverse effects

- Not available

13. DISPOSAL CONSIDERATIONS

A. Disposal methods

- Since more than two kinds of designaed waste is mixed, it is difficult to treat seperatly, then can be reduction or stabilization by incineration or similar process.
- If water separation is possible, pre-process with Water separation process.
- Dispose by incineration.

B. Special precautions for disposal

- The user of this product must disposal by oneself or entrust to waste disposer or person who other's waste recycle and dispose, person who establish and operate waste disposal facilities.
- Dispose of waste in accordance with all applicable laws and regulations.

14. TRANSPORT INFORMATION

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A. UN No. (IMDG)

- 1919

B. Proper shipping name

- METHYL ACRYLATE, STABILIZED

C. Hazard Class

- 3

D. IMDG Packing group

- II

E. Marine pollutant

- Not available
- Not applicable

F. Special precautions for user related to transport or transportation measures

- Local transport follows in accordance with Dangerous goods Safety Management Law.
- Package and transport follow in accordance with Department of Transportation (DOT) and other regulatory agency requirements.
- Air transport(IATA): Not subject to LATA regulations.
- EmS FIRE SCHEDULE : F-E (Non-water-reactive flammable liquids)
- EmS SPILLAGE SCHEDULE : S-D (Flammable liquids)
- Self Accelerating Polymerization Temperature (SAPT) : 60℃
- This product is stabilized by addition of polymerization inhibitor (MEHQ).

15. REGULATORY INFORMATION

A. National and/or international regulatory information

○ POPs Management Law

- Not applicable

○ Information of EU Classification

* Classification

- [2-Propenoic acid methyl ester] : H225,H332,H312,H302,H319,H335,H315,H317

○ U.S. Federal regulations

* OSHA PROCESS SAFETY (29CFR1910.119)
  - Not applicable
* CERCLA Section 103 (40CFR302.4)
  - Not applicable
* EPCRA Section 302 (40CFR355.30)
  - Not applicable
* EPCRA Section 304 (40CFR355.40)
  - Not applicable
* EPCRA Section 313 (40CFR372.65)
  - [2-Propenoic acid methyl ester] : Applicable

○ Rotterdam Convention listed ingredients

- Not applicable

○ Stockholm Convention listed ingredients

- Not applicable

○ Montreal Protocol listed ingredients

- Not applicable

16. OTHER INFORMATION

A. Reference

- The information contained herein is believed to be accurate. It is provided independently of any sale of the product for purpose of hazard communication. It is not intended to constitute performance information concerning the product. No express warranty, or implied warranty of merchantability or fitness for a particular purpose is made with respect to the product or the information contained herein.
- This Safety Data Sheet was compiled with data and information from the following sources: KOSHA, NITE, ESIS, NLM, SIDS, IPCS

B. Issue date

- 2017-11-22

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C. Revision number and Last date revised

- 12 times, 2017-11-22

D. Other

- This SDS is prepared according to the Globally Harmonized System (GHS).